REVIEW ANALYSIS AND EVALUATION—DIAGNOSIS/TREATMENT/PROGNOSIS

INTRA-ARTICULAR PHARMACOLOGICAL INJECTIONS FOR TEMPOROMANDIBULAR JOINT OSTEOARTHRITIS ARE COMPARABLE TO PLACEBO

PURPOSE/QUESTION In adult patients with temporomandibular joint osteoarthritis (TMJ OA), is the intra-articular injection of either corticosteroids (CCS), hyaluronic acid (HA), platelet-rich plasma (PRP), or placebo effective in reducing joint pain?

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Article Title and Bibliographic Information
Effectiveness of intra-articular injections of sodium hyaluronate, corticosteroids, platelet-rich plasma on temporomandibular joint osteoarthritis: a systematic review and network meta-analysis of randomized controlled trials. Xie Y, Zhao K, Ye G, Yao X, Yu M, Ouyang H. J Evid Based Dent Pract. 2022 Sep;22(3):101720. doi:10.1016/j.jebdp.2022.101720.

SORT SCORE

A	B	C	N/A

B (selected)

SORT, Strength of Recommendation Taxonomy.

LEVEL OF EVIDENCE

1	2	3

2 (selected)

See article 101482 for complete details regarding SORT and LEVEL OF EVIDENCE grading system.

SUMMARY

Subjects or Study Selection
PubMed, Embase, and the Cochrane Central Register of Controlled trials were searched for randomized controlled trials on therapies for temporomandibular joint osteoarthritis. The gray literature was searched via ClinicalTrials.gov. Reference lists of reviews were also manually scanned. Two reviewers independently assessed the risk of bias using the dimensions of quality in randomized controlled trials.

Key Study Factor
Included studies compared various combinations of intra-articular injections of either corticosteroids (CCS), sodium hyaluronate (HA), plate-rich plasma (PRP), or placebo for temporomandibular joint osteoarthritis (TMJ OA).

Main Outcome Measure
The main outcome was pain on a visual analog scale (VAS) 1-10 after short-term (3-6 months) or long-term (>12 months) follow-up. Scales from 0 to 100 were converted to a 0-10 scale.

Main Results
Pain was reported as a mean difference with 95% CI. The bulk of results were *not* statistically significant with one notable exception: for short-term pain reduction, injection of CCS was significantly worse than placebo (MD = 1.73; 95% CI 0.37-3.09). The abstract reported PRP vs placebo as MD = −0.23 (95% CI −2.49 to 2.04), one of the many nonsignificant results.

SOURCE OF FUNDING
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TYPE OF STUDY/DESIGN
Systematic review with network meta-analysis of data.

KEYWORDS
Temporomandibular disorders, Osteoarthritis, Sodium hyaluronate, Platelet-rich plasma, Corticosteroids

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Conclusions

Intra-articular injections of either CCS, HA, or PRP for TMJ OA had no effect on improving pain reduction compared with placebo.

COMMENTARY AND ANALYSIS

The American Academy of Orofacial Pain (AAOP) defines TMDs as a group of musculoskeletal and neuromuscular conditions that involve the temporomandibular joints, the masticatory muscles, and all associated tissues.[1] Thus, TMDs encompass a number of different diagnoses and are a major source of non-dental pain and/or dysfunction in the orofacial region.[2-4] TMDs are generally acute, but can become chronic resulting in significant morbidity and quality of life issues. There are now several clinical practice guidelines on the management of temporomandibular disorders and they resonate with each other in that many TMDs have spontaneous remissions amenable to conservative, reversible therapies. Conservative therapies would include the use of analgesics, heat application, and physical therapy. A recent report by the National Academies of Sciences, Engineering, and Medicine (NASEM) noted a number of deficiencies regarding the education of TMDs leading to inconsistencies with both diagnosis and treatment.[5] While TMJ OA has been treated by intra-articular injections of various pharmaceuticals, studies on the effectiveness of these medications versus placebo (Ringer's Lactated Solution) have been lacking. Thus, this network meta-analysis is clinically relevant. Another important finding is the lack of statistical significance of these injectables as compared with placebo. However, it must be noted that the 9 included studies involving 316 patients were all generally small, with only 1 study having 102 patients. This large included study showed that CCS injections were significantly *worse* than placebo. Generally, we have non-significant results with multiple underpowered studies which leaves the effectiveness of intra-articular injections with a degree of uncertainty. That is, with additional studies hopefully having larger samples, would summary estimates move away from the null and show a benefit to these therapies, or might they simply confirm that there is no effect? Either way, additional studies should narrow confidence intervals providing stronger evidence to answer this important question. In addition to the imprecision (ie, a wider confidence intervals) of results due to small samples, there was also considerable inconsistency (heterogeneity) with some results on opposite side of the null value.

Methodologically, this was a very well-conducted network meta-analysis (MA), and the authors are to be commended on this work. A network meta-analysis has the direct comparisons of traditional MA, and also allows for indirect comparisons not done in the included studies, especially enabling comparisons with placebo injections.

In addition to multiple small studies, some of the included studies were at unclear risk of bias, emphasizing the need for larger, high-quality randomized trials to help answer this important clinical question. Therefore, clinicians may want to use caution with intra-articular injections for TMJ OA as the current best evidence fails to show a benefit of various injectables as compared with placebo.

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5. National Academies of Sciences, Engineering, and Medicine. *Temporomandibular Disorders: Priorities for Research and Care*. Washington, DC: The National Academies Press; 2020.

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